

Knowledge construction among the Cree hunters : metaphors and literal understanding

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Résumé

La construction du savoir chez les chasseurs Crée : métaphores et compréhension littérale. Il y a longtemps que l'anthropologie insiste sur les aspects magiques et mythiques du savoir dans les sociétés non-occidentales, en les opposant à la rationalité réputée littérale et empirique de la science occidentale. Cette opposition survit dans la théorie contemporaine de la culture, quoique sous des formes indirectes. Cet article examine le savoir cynégétique des Crée, et montre que le mythe et la mise en forme littérale de l'expérience empirique sont des moments d'une transformation dans la construction du savoir. Aucun de ces moments, à proprement parler, n'est privilégié ni indépendant ; les prémisses de la communication et de la réciprocité entre humains et animaux engendrent un savoir empiriquement sophistiqué.

Abstract

Anthropology has long emphasized the mythico-magical aspect of knowledge in non-western societies, by contrast to the supposed literal-empirical rationality of western science. The opposition survives in contemporary cultural theory, albeit in indirect forms. The present paper examines Cree hunting knowledge, to show that myth and the literal modelling of empirical experience are mutually transformative moments in the construction of knowledge ; that neither moment, properly speaking, is privileged or independent ; and that premises of human/animal communication and reciprocity are productive of empirically sophisticated knowledge.

Resumen

La construcción del saber entre los cazadores Cree : metáforas y comprensión literal. Hace tiempo que la antropología insiste sobre los aspectos mágicos y míticos del saber en las sociedades no-occidentales, al que opone la racionalidad, considerada literal y empírica, de la ciencia occidental. Esta oposición sobrevive, bien que de manera indirecta, en la teoría contemporánea de la cultura. Este artículo analiza el saber cinegético de los Cree y muestra que tanto el mito como la elaboración de la forma literal de la experiencia empírica son momentos de una transformación en la construcción del saber. Ninguno de esos momentos es privilegiado o independiente ; las premisas de la comunicación y de la reciprocidad entre humanos y animales crean un saber empíricamente sofisticado.

KNOWLEDGE CONSTRUCTION AMONG CREE HUNTERS : METAPHORS AND LITERAL UNDERSTANDING

Colin SCOTT *

Anthropology has long emphasized the mythico-magical aspect of knowledge in non-western societies, by contrast to the supposed literal-empirical rationality of western science. The opposition survives in contemporary cultural theory, albeit in indirect forms. The present paper examines Cree hunting knowledge, to show that myth and the literal modelling of empirical experience are mutually transformative moments in the construction of knowledge; that neither moment, properly speaking, is privileged or independent; and that premises of human/animal communication and reciprocity are productive of empirically sophisticated knowledge.

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Hace tiempo que la antropología insiste sobre los aspectos mágicos y míticos del saber en las sociedades no-occidentales, al que opone la racionalidad, considerada literal y empírica, de la ciencia occidental. Esta oposición sobrevive, bien que de manera indirecta, en la teoría contemporánea de la cultura. Este artículo analiza el saber cinegético de los Cree y muestra que tanto el mito como la elaboración de la forma literal de la experiencia empírica son momentos de una transformación en la construcción del saber. Ninguno de esos momentos es privilegiado o independiente; las premisas de la comunicación y de la reciprocidad entre humanos y animales crean un saber empíricamente sofisticado.

La construction du savoir chez les chasseurs Cree : métaphores et compréhension littérale.

Il y a longtemps que l'anthropologie insiste sur les aspects magiques et mythiques du savoir dans les sociétés non-occidentales, en les opposant à la rationalité réputée littérale et empirique de la science occidentale. Cette opposition survit dans la théorie contemporaine de la culture, quoique sous des formes indirectes. Cet article examine le savoir cynégétique des Cree, et montre que le mythe et la mise en forme littérale de l'expérience empirique sont des moments d'une transformation dans la construction du savoir. Aucun de ces moments, à proprement parler, n'est privilégié ni indépendant; les prémisses de la communication et de la réciprocité entre humains et animaux engendrent un savoir empiriquement sophistiqué.

The present analysis explores the complementarity of figurative and literal signification in knowledge construction among eastern Cree hunters. The examina-

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tion focuses on the way in which root metaphors of pan-species *personhood*, *communication*, and *reciprocity* inform literal models of animal behaviour and hunting practice; and how the latter reciprocally transmute the terms of metaphor, as experience is interpreted and actions are formulated.

Anthropology has yet to dissolve the evolutionary opposition between the science of "the west" and the myth/magic of "the rest". Ethnoscience fieldwork of the 1960's turned up empirically elaborate classifications and orders of classifications from a wide range of "traditional" societies. For Lévi-Strauss (1966), these empirical categories, or "percepts", were signs in the *bricolage* of mythical thought. The structures of mythical thought often produce scientific results because the mind, perception, and the external world share a common "natural" foundation. According to Lévi-Strauss (and Evans-Pritchard before him) the structures of reason in myth and magic were not fundamentally different from those of science. Yet in Lévi-Strauss, the science/myth opposition is salvaged via the curious notion that mythical "signs" are somehow corralled by their empirical signifiers, while the "concepts" of science are more free (*ibid.* : 18-20).

Reichel-Dolmatoff (1976) demonstrated the formal compatibility between mythico-ritual structures and ecosystemic principles among Amazonian Tukano. And Rappaport's work (1979) on Maring cosmology has shown that supernatural abstract categories have homeostatic functions in highland New Guinea ecosystems. For both these writers, however, the structural or functional connections between abstract cosmology and ecosystems are constructs of the analysts. This is particularly explicit in Rappaport's functionalist articulation of an emic "cognized model" to an etic "operational model" of the ecosystem. The operation of cosmology as practical knowledge, from the actors' point of view, remains largely hidden.

What, then, has been overlooked? With a few exceptions (e.g. Feit 1978, Nelson 1983), anthropologists have not examined how mythico-ritual categories are implicated in hunters' literal modelling of social-environmental practice; in situational, strategic discourse about material activity. In place of actors' models, we have substituted our « etic » ones. The adaptation of native cosmologies to their material-historical environments therefore appears fortuitously functional, or a happy congruence of symbolic and material structures (if the congruence is to be believed), rather than the outcome of theoretical work on the actors' parts.

Here I will proceed in three steps : 1. to consider practical experience as it signifies within a Cree cosmology and epistemology; 2. to consider a useful semiotic model of literal/figurative complementarity; and 3. to turn to examples from Cree goose hunting that show alternation between literal and figurative modes of interpreting environment, each providing context and definition to the other.

Signs in Cree Epistemology.

In the standard Euro-North American perception of human/animal relations, culture/nature dualism is fundamental. It is true we have begun to culturalize animals in animal communications studies, and to naturalize culture in anthropological ecology. But our conservatism is to assume fundamental differences, while

working out the similarities. The Cree trajectory seems rather the opposite : to assume fundamental similarity while exploring the differences between humans and animals.

In Cree, there is no word corresponding to our term 'nature'. There is a word *pimaatisiwin*, 'life', which includes human as well as animal 'persons'. The word for 'persons' *iyyiyuu*, can itself be glossed as 'he lives'. Humans, animals, spirits and some geophysical agents are perceived to have qualities of personhood. All persons act in a reciprocally communicative reality. Human persons are not set over and against a material context of inert nature, but rather are one species of person in a network of reciprocating persons. These reciprocative interactions constitute the events of experience.

Again, there is no Cree equivalent of 'culture' that would make it the special province of humans. Crees do, however, have terms which resemble the semiotic notions of 'sign vehicle' (*chischinawaachaawapihtawaawan*) and the 'meaning/interpretant' (*iishchiishwaamaakan*) of the sign. Cognates of these terms, incorporating the morpheme *-chis-*, acknowledge the ubiquity of signs in experience. They include *chischaaimaau*, 'he knows (him)'; *chischaaitemuun*, 'information, knowledge'; *chischinutihaau*, 'he leads, directs, guides (him)'; *chischinuwaasinaakusuu*, 'he is used for a sign/he gives a sign'; and *chischiwaahiicheu*, 'he prophecies' (Mackenzie 1982). *Achiskuhiichaawan* and *achiskuhiikan* refer to markers left in the bush to indicate to others where one's path leads, how far, etc. Animal behaviour, dream images, and religious symbols all fall within the Cree notion of 'sign', with the assumption that signs constitute knowledge or guidance for actors. And not only humans, but animals and other non-human persons send, interpret, and respond to signs pertinent to various domains of human action : hunting success or failure, birth and death and, critically related to these, the circumstances of reciprocity between persons in the world.

Signs, then, are part and parcel of action, perception and experience — of life itself. The term *pimaatisiwin*, 'life', was translated by one Cree man as 'continuous birth'. Consciousness (*umituunaaichikanich*, glossed by the same man as 'mind and heart, thought and feeling') is on the verge of unfolding events, of continuous birth. There are many signs of recurrence and regularity in experience, but interpretations are seldom certain or absolute. To make fixed plans on the basis of predictions about the future, for example, may presume too much. Someone (human, animal, or spirit) could reciprocate by frustrating hunters' intentions.

Literal/figurative Complementarity.

Since events are the efflux of actions of many mutually responsive persons, human action is subject simultaneously to moral and technical criteria of evaluation. This is a commonly remarked feature of worldview among huntig and other "traditional" societies. Roy Wagner (1977, 1981) attributes it to a predominance of figurative ("differentiating") signification in traditional societies, by contrast to the literal ("collectivizing" & "relational") bent of signification in Western culture. The opposition is overdrawn as one of societal types, but the grounds on which it is presented are illuminating for the present discussion.

Literal and figurative constructs are complementary, mutually dependent aspects of all signification. A construction either relates the perceptibly differentiated, or differentiates the perceptibly relational. To take a simple example, "natural" objects (and "nature" assumed as an objective context) are the largely assumed figurative *differentiae* of western science. Literal, deliberate cause-effect constructions which relate objects are possible only because the objects are assumed to begin with. Literal signification (e.g. scientific culture) combines objects into total relational patterns which represent a context (maximally, "nature"). Natural objects, implicitly assumed, seem to be innate, and nature as represented context is set apart or differentiated from its cultural representation. Figurative signification, on the other hand, is explicitly concerned with constructing similarities and differences between semiotic elements or "objects", via analogy between objects or entities in the world, and assumes certain relational contexts (e.g. relations of reciprocity among some non-western people).

Nature and natural phenomena, then, are the (implicit) figurative counterparts of Western literal science. If we accept literal construction as our chief business, we accept the separation of culture from nature as quite "natural". But a society which concerns itself with analogies between phenomena in the world and itself will precipitate its own conventional context (e.g. communicative reciprocity) as the innate character of phenomena in general.

Since literal signification relates *differentiae* in a "represented" context, while figurative signification differentiates elements in relational context, they are simply two faces of any knowledge construct. This combats the tendency to see myth and magic as structurally separate from science in a given cultural tradition. All signs, from the abstract and cosmological to the empirically discrete, can be assessed for both literal and figurative significance.

I will argue that it is alternation between figurative and literal modes, not emphasis on one or the other, that leads to the integration of moral and technical criteria in the ecological practice of Cree hunters. The Pacific cultures discussed by Wagner, much like Cree, view the world as an innate realization of a conventional social order of reciprocity, or so it seems if we take Cree statements of cosmology at the face value perceived by a Westerner. In Cree constructions, as we have noted, communicative exchange is extended so ubiquitously to non-human domains that it constitutes a "root metaphor" (Ortner 1973) or "paradigm" (Black 1962) for knowledge in general. Myth, ritual, dreams and hunting scenarios all express respectful solicitude as the preferred relation among "persons" in the hunter's world. The mental and physical activity of the hunter is directed at maintaining standing in this network, by being generous and respectful to humans and non-humans, and by ensuring that what is received is in correct proportion. Where moral standards of positive reciprocity are deviated from, its negative corollary ensues : in all contexts, generosity dwindles in response to disrespect and greed.

But the viability of these common premises depends on the rigorous construction of differences between persons in every instance. In common with many cultural traditions,

It is by maintaining a precise awareness of these differences, by differentiating himself and by differentiating the various beings in an appropriate manner, that man precipitates (or from the actor's point of view, invokes) a beneficent relational flow (Wagner 1977 : 398).

The permutations of communicative reciprocity as a root metaphor vary with the phenomena which are its signifiers in various literal contexts.

Knowledge Construction in Cree Goose Hunting.

Goose hunting is a major ritual and economic event during the exceptionally heavy fall and spring migrations along the coasts of Hudson and James Bays. In certain sacred images, the identities of hunters and animals are so passionately condensed that the aspect of similarity in root metaphor virtually eclipses the aspect of difference. But in most practical contexts, the differences between hunters and animals as reciprocating agents are in the foreground. The definition of these differences flows from quite deliberate relational models which connect hunters and game metonymically — in consumer/consumed complementarity and in cause-effect orders which include other environmental agents such as winds, tides and topographical features.

Reciprocity among humans is constrained differently from reciprocity between humans and animals. In the first instance, biological structures of human reproduction signify a fundamental separation and asymmetry between human community and animal community, in respect of the former consuming the latter. The justification of this asymmetry is no trivial matter. Several Cree myths are concerned with human sexuality as a metaphor the killing and eating of game, and *vice versa*. When humans get the terms of their metaphors confused and begin marrying animals or eating other humans (i.e. fail to differentiate correctly), the results can be impossible or tragic (Preston 1975, 1978). That an animal be available for human consumption is an index of respect and love between hunter and animal. But it is horrifying to contemplate the consumption of other humans.

The metaphoric juxtaposition/separation of humans and animals is the occasion for much humorous discourse linking the pursuit of sexual partners to the pursuit of game. Hunting and sexuality share a vocabulary : *mitwaaschaau* can mean both 'he shoots' and 'he ejaculates'; *paaschikan* can refer to both 'shotgun' and 'penis'; *pukw* to both 'gunpowder' and 'sperm'; and *spichinaakin* to both 'gun sheath' and 'condom'.

The *atuush*, or 'cannibal' subverts the separation of human and animal persons. In one bawdy myth, a cannibal copulates with a woman hunted by his son, before roasting and eating her reproductive organs. In consequence, he consumes his own sperm. He and his son, greatly weakened, are nearly overcome by the spiritual power of true human beings.

Another myth illustrates the necessity of killing animals, while respecting certain parameters for doing so. It concerns a supernatural character, *Chischihp*, who never ate. *Chischihp* thinks of the food animals whom he loves as his 'pets', or 'dogs'. In this he is opposed to Cree hunters. Hunters also refer affectionately to certain species as their 'pets', but normally it would be the species that an individual hunter is privileged to kill with unusual success. *Chischihp* would never have begun killing his 'pets' had he not met two human sisters on a river journey,

and desired them for wives. They accept his proposal, but insist that he kill beaver and moose for them, and when he objects, they threaten to abandon him. He relents, kills the animals, and eventually begins to eat some of the meat himself. However, he goes from excessive abstinence to excessive indulgence, both in respect of the animals and of the women, following his conversion to human status. When he returns to his village, he selfishly hides his wives, preventing their attendance at a public dance. At the dance, he himself dons the fatty internal organs and membranes of the moose as clothing and personal adornment. These parts are esteemed food delicacies, and their ostentatious display is grossly disrespectful of the animal. His wives respond with infidelity. He discovers them sleeping with a close agnate, whom he promptly murders in vengeance. He is now classified as a *pwaat*, a sub-human person who lurks at the margins of true human community and who shares some attributes of cannibals. Through treachery, he escapes the wrath of his village, drowns his wives, and is himself transformed into a species of edible waterfowl, the form in which he is known to Cree hunters today. Human reproduction, then, demands the consumption of animals as positively as it prohibits the consumption of other humans; but there are respectful parameters for both inter-species consumption and intra-species sexuality that are specific to the form of "reciprocity" in question.

I want to go on now to illustrate that the literal representation of animal behaviour in environmental context impels the further figurative differentiation of human and non-human persons. The Cree make ample use of humans and animals as interpretants of one another. The family structure, leadership, memory and communication processes of geese are explored as analogues of the corresponding human qualities, both individual and social. I will focus on goose communication and learning, and associated hunting strategies, in the current example. Geese as objects of knowledge are extremely important both ritually and practically they account for as much as one-quarter of all annual subsistence production for coastal Cree.

One might observe that a consequence of the sort of analogical thinking that I have been describing would be to anthropomorphize animals, but that would be to assume the primacy of the human term. The animal term reacts with perhaps equal force on the human term, so that animal behaviour can become a model for human relations. Preston (1978 : 152) has suggested that the goose as exemplar of Cree ideals of social coordination, grace and composure may be "better" than humans.

There are also heuristic advantages to the Cree paradigm of a sentient, communicative world transcending humanity. This paradigm has oriented Cree to aspects of animal behaviour that Western science, inured to a Cartesian metaphor of mechanical nature, has admitted rather belatedly. Lorenz (1979) observes that for the "higher" animals, the expression of emotion involves substantially the same neuro-muscular system as in humans. Geese possess some quite "human" affective qualities, including loyalty, jealousy and grief; furthermore, these qualities are manifest in the context of striking similarities in courtship, mating and the rearing of young. "The family and social life of wild geese exhibits an enormous number of striking parallels with human behaviour", Lorenz (*ibid.* : 192) observes; "Let no one think it is misleading anthropomorphism to say so".

Lorenz finds a greater gulf separating the rational faculties of geese and humans. Yet here, too, ethologists are finding that animals classify elements of environment with some sophistication :

Animals create a taxonomy appropriate to their species and ecological niche. Thus predators, for instance, distinguish different categories of prey — by size, appearance, odor, and other signifiers — thus forestalling wastefully indiscriminate attacks. *Vice-versa*, many potential prey distinguish among different kinds of predators as we observe from their use of sundry warning signs, variations in their flight-distances and flight-reactions (Sebeok 1975 : 62).

Sebeok has argued that for animals, as for humans, the aesthetics are intimately linked to the extraction and reconstruction of structures from salient environmental features, “even when the process or the product is disunited from its proper biological context » (*ibid.* : 61).

The perceptions and interpretations of Cree hunters suggest that geese are quite apt at learning in what contexts to expect predation, at learning to distinguish predatory from non-predatory humans, and at communicating appropriate behavioural adaptations to other geese. In other words, there is substantial flexibility for geese to “reinterpret” environmental signs, and this learning is communicated among geese to ramify socially.

It is therefore important for hunters to arrive at precise estimations of goose learning and communication, particularly in relation to themselves as predators. It is heuristically useful but not in itself sufficient to assume, on the basis of the culturally pervasive paradigm, that capacities of intelligence and communication are shared by geese and humans. Hunters need to know about both similarities and differences. The more the respective capacities of geese and humans are specifically formulated (i.e. differentiated), the more “literally” they contribute to effective hunting scenarios. The interpretation/modelling of hunting experience is an ongoing refinement of hunters’ knowledge of the specific capacities of geese, leading to adjustments in hunting practice.

Hunters arrange landscapes that will be attractive and non-threatening to geese, while exercising caution so that geese will not learn to associate unusual details with the possible presence of hunters. Decoys and and goose calls are iconic approximations by hunters of the semiotic landscape of geese. Hunters recognize variation between species in regard to the features which geese notice. Canada goose decoys must be realistic in profile and must be kept pointing into the wind, properly spaced, with decoys in both feeding and alert postures. Generally speaking, greater numbers of decoys are more effective. Snow geese are less sensitive to profile or number of decoys, but respond strongly to colour. Two or three white plastic buckets or white rags displayed prominently on a hillside, in conjunction with calling, are sufficient.

The repetitive honking of the Canada goose is imitated to get the attention of an approaching flock once it is near enough to spot decoys. When the geese have seen the decoys and “made up their minds” to fly over to land, hunters stop calling, or switch to two or three long, low “welcoming” calls at gentle intervals. Calling should be used sparingly — novice hunters must learn both to imitate goose calls accurately, and to know when *not* to call. If a flock in the air has

chosen not to respond to calls but to continue on and away from blinds and decoys, hunters should cease calling, because geese will only recognize that such calls are unnatural. When there is less wind, the geese hear calls (and mistakes) more keenly. When a hunter is especially skilled at calling geese, others may prefer to keep silent to reduce the risk of detection.

Neither the semiotic conventions of geese, nor their interpretation and manipulation by hunters, are static. Geese, like hunters, are said to "know the land". Their ability to recontextualize certain perceptual features as signifying the presence of hunters is the potential undoing of the latter, as geese "get wise". For this reason, hunters' precautions to minimize visual and auditory signs of their own presence go well beyond the use of blinds at actual hunting spots. Camps are kept at some distance from concentrations of geese, and are well-hidden in the bush. Snowmobiles and chainsaws are not used near concentrations of geese. Ideally, the only birds on the territory that will be immediately aware of hunters' presence are those from small flocks actually fired on at hunting sites. Shooting on calm days is generally avoided, because the sound of shooting carries over a wide area without a wind to muffle and disperse it. Shooting after dusk is also avoided, because the flame visible at night at the end of a fired shotgun is said to terrify geese. Similarly, the use of lights outdoors at night is restricted.

Hunters' experience is that geese will not return to a hunting spot that has been used too regularly, or where they have been frightened badly. The Wemindji community area along the coast of James Bay is divided into several goose hunting territories, each used by up to a dozen hunters from a number of households linked agnatically, affinally, and by friendship. Hunting activities for each group are under the supervision of a senior "shooting boss". Each territory includes a number of viable hunting spots, and all hunters on a territory are expected to use one and the same spot on any given day, allowing all other spots to "rest". Normally, a new site is chosen each day, so that hunting spots are rotated. In this way, the migrating geese will not learn to expect hunters at any particular location, and will be respectfully permitted to rest and feed undisturbed over the majority of the territory on any given day. At hunting sites, when geese are killed, it should be done accurately and efficiently, to minimize disturbance and to avoid waste of injured birds that escape.

If these precautions are not taken, geese on the territory will grow increasingly anxious about human presence and adjust their behaviour accordingly. Even a fraction of geese too badly frightened communicate their alarm to other geese, which could lead to a reduction in the population staying on the territory, or to incremental avoidance by geese of the spot where the fright occurred. Since the same geese are on the migration route in successive seasons, and since young geese are said to learn their habits from their parents, a hunting spot which has been mishandled can take several seasons to recover.

Here I will limit the discussion to one example of changes in goose behaviour that are attributed to hunting. In early autumn, when the tide is high and the wind is brisk and onshore from James Bay, the geese fly from the coastal bays to the offshore islands in the Bay, early in the morning. They do so to feed on berries there, partly because high tides, onshore wind and rough waves make it impossible to feed on the eelgrass which grows in shallow water in the coastal bays. The

berries, now ripened, are much sought-after by the geese, besides. This relational model is of key importance in hunters' decisions about where and when to locate themselves to wait for geese. When experience fails to confirm the expected relations, amendments to the model ensue.

It has in fact developed in recent years that geese are less prone than before to fly to hunters waiting on the islands, even when there have been plenty of geese in the coastal bays, and circumstances of wind and tide have been favourable. The interpretation of this decline by experienced hunters is as follows : In recent years, population growth and wage and transfer payment income have led to an increase in the number and mobility of hunters who have greater access to motorized watercraft. This means that any hunting location in the coastal community area is generally accessible within twenty minutes to two hours' travel from the settlement.

Significantly, these settlement-based hunters do not independently enter the more sensitive coastal bays, where concentrations of geese rest at night and also feed when the tide is low. Only when such a bay is being hunted as part of a rotational strategy, under the leadership of a 'shooting boss', are hunters from the settlement welcome to join. But offshore islands can be used with much less risk of disturbance to the main concentrations of geese in the bays, particularly when there is enough wind to prevent the sound of shooting from carrying far. It has developed as a sort of community compromise between full-time hunters based in camps and wage-earning part-time hunters, who are less flexible as to the times they can hunt, that the islands can be used without direct supervision and coordination by shooting bosses. This allows settlement-based wage-earners to hunt on days-off or after-hours, even when conditions are not suitable for hunting in the bays.

However, because this hunting at the islands has become more frequent and is no longer coordinated in regular rotation, the geese have come increasingly to expect hunters at the islands. Consequently, there have been more geese flying inland instead of offshore, when they leave the bays to feed. Hunting geese inland is less productive and more difficult of access, so that geese have learned not to expect hunters there. They are observed to fly lower and less cautiously inland than along the coast and offshore islands.

The kind of interpretation just summarized, and the hunting strategies entailed, involve years and generations of practical empirical investigation into how much and what kinds of interactions with hunters the geese will tolerate, without withdrawing from strategic locations. The relational models conventionally signified by this experience help to define goose communication as different in degrees and respects from human communication. Literal constructions precipitate a figurative complement — the differentiation of human from goose communication, or of goose communication from that of some other animals.

Some of the attributes of goose communication itself are best contemplated anecdotally. Here, metaphors that evoke human leadership, speech, etc., are evident, but the hunting situations referred to are themselves key interpretants of the appropriate extent and application of the metaphor. A friend and I were sitting in our blinds early one spring. A few larger flocks were heading north, but they were too high and were not coming into our calls and decoys. There was a lake

nearby, perhaps a half kilometer back into the bush but within earshot. We could hear the gabbling of numerous geese there. They hardly flew all day, until in mid-afternoon one solitary member came low over the trees toward us. It wheeled to land among the decoys, an easy shot. We fired half a dozen shots, somehow missing, and the goose fled back in the direction of the lake. My companion speculated on the consequence of our poor shooting, a few moments later : "Probably that goose told the other ones over there: 'If I don't come back, it's okay to come on over'". No more geese flew our way from the lake that day. I didn't think that my companion intended that geese are literally possessed of speech. But it was clearly plausible to him that geese communicate to other geese about phenomena the latter have not experienced directly. "Scouting" among geese is observed in a variety of contexts, and presumably the behaviour of these scouts conveys something about attractive versus dangerous situations. A variety of calls and postures, in flight and on the ground, are distinguished by Cree hunters : from messages of invitation to those of caution and alarm.

On another occasion later the same spring, the same companion and I were waiting for geese on an east-west elevated ridge between two coastal bays. The geese, who were leaving the bay to the south to continue their migration northward, would fly over the ridge. The ridge was perhaps two hundred meters in length along its bare peak, affording hunters view of approaching flocks. It happened several times that a flock would cross too far east or too far west of our spot, so we would be unable to get a shot. This seemed random enough, but then one or more flocks following at intervals of several hundred meters would cross at the same spot as the first flock, again evading us. I wondered if winds might account for this regularity, but rejected that possibility because the flocks had crossed at the same points even when they had approached the ridge from quite different trajectories. Then too, the occasional flock would fly directly over us. I was ready to attribute the pattern of evasion to episodes of unfortunate coincidence until my companion remarked : "It always seems to happen like that. I guess they know there's hunters around. They see where the flock ahead of them went over, and they see nothing happens to them, so they think, ' Might as well go over there!'".

This incident illustrates the attentiveness of even separate flocks to the activity of other flocks in selecting a course of action. The safe patterns of a few geese may be copied by many, in a context where mature geese have learned to expect hunters. It became more plausible to me that avoidance of certain situations and preference for others could ramify socially among geese, resulting in general behavioural changes for the population as a whole.

At the same time, several aspects of goose awareness and communication remain esoteric. The capacity of animals to anticipate some events is considered superior to that of humans. Goose behaviour of certain kinds is a predictor of approaching weather; geese begin their preparations before hunters would otherwise be aware of impending changes. Another example is that in years when the local berry crop has failed, most geese on the fall migration have been observed to fly very high right on past, not even landing along the James Bay coast. What is outstanding to hunters is that the flocks do not have to land to know that the feeding is poor. One interpretation of this phenomenon relates again to scouting

behaviour, although the precise mechanism of information transfer is ambiguous. In some interpretations, it is supposed that animals experience counterparts of dream images and corporeal symptoms which alert humans to future or distant events. These are notable but not particularly unusual experiences in the Cree world.

This common denominator returns us to the premise of a communicative, reciprocative network which unifies the holistic world. This premise is metaphysically prior to the differentiation of persons in the world — and it is at this level that hunters, animals, geophysical forces, and even God are ultimately of one mind, as it were. Consistent with this premise is the notion that encounters, thoughts, dreams, and rituals involving hunters-animal exchange both index and influence the state of reciprocity which obtains at a given point in time.

Animal and Human Reciprocators.

A previous section has discussed the operation of metaphors of eating and sexuality in establishing a fundamental difference between human-human and animal-human reciprocity. I'll go on now to illustrate that ritual delineations of reciprocity between humans and animals contain, at the same time, abstract but quite literal constructions of social-ecological principles. The life cycle of the hunter, seasonal cycles, the social roles of men and women are all marked by ritual stages which reflect these principles. A few examples will suffice.

Early in the spring season, the geese killed on the first day of the hunt are cooked and a small share is distributed by a respected woman elder to everyone in a camp. This is done no matter how few the geese or how large the camp, and regardless of who killed the geese. Then, geese are saved for a few days until there are enough that every man, woman and child in the hunting group receives one, two or more geese, depending on how many have been killed. Again, the distribution is made by a respected elder of the camp, and without regard to who killed the geese. The group then feasts, with each household roasting some of the geese it has received. This process is considered to be an "invitation" to the geese, since animals are said to come more readily to hunters who share them with others. Only after this feast is it possible for the household to accumulate geese for its own consumption. Significantly, the feast occurs while it is still to be determined whether the migration will linger, ensuring bounty, or pass quickly. Failure to contribute generously to the feast can be used to account for poor luck in hunting later on.

A hunter in his blind often 'smokes to the game' (*pwaatikswaau*), or sings goose songs. Tobacco was a traditionally valued item of exchange, and smoke is an appropriate vehicle of exchange with creatures of the air. Hunters' songs express vivid images of the ways in which geese fall to the hunter. When a goose has fallen, the gift is respectfully admired by the hunter, and later received as a guest into the lodge, by the women of the hunter's household. The women take care to use every part of the goose possible, to avoid spoilage, and to dispose respectfully of the few remains. The cartilage tracheae, including the windpipe and voice organ of the goose, are hung from a tree branch where, poetically, the passing wind carries their call, beckoning geese in future seasons to renew the exchange¹. When the

migration has nearly passed and the last of the geese are departing, the hunter bids them farewell, expressing the hope that, granted continued life, he will be able to see them again on their return.

The ritual complex advances two general propositions about human-animal reciprocity, of key ecological concern. The first is that respectful activity towards the animals enhances the readiness with which they give themselves, or are given by God, to hunters. The second is that sharing of animal gifts among hunters is an important dimension of respect for the animals. Both propositions are implicit in ritual enactments of the special obligations of hunters toward game, if game animals are to perform their own special role in supplying hunters. And both propositions convey literally-understood truths about ecological relations².

Let us deal at this point with the first of these general propositions. The empirical availability of geese, as we have seen, varies with their treatment by hunters. The specification of "respectful" treatment in day-to-day hunting is as complex as the many situations of interaction, but the general and key notion is that technical efficiency in killing animals must be balanced by restraint, and that only the latter can really guarantee the long-term viability of the former.

A hunter must strive for impeccable technique, both in the interest of his own security and to avoid undue suffering or disturbance to the animals. A hunter who in spite of his effort cannot kill animals will be disappointed, but he should not be too disappointed because it is wrong to expect more than is offered freely by a partner in reciprocity. The hunter may not be receiving more because the partner cannot, consistent with its own welfare, offer more. The wise hunter directs his efforts elsewhere, if after trying hard it is apparent that a particular species doesn't want to be caught. On the other hand, it is wrong to accept more than one needs, even if it can be taken with the means at one's disposal. The generosity of a partner can be overtaxed.

There is a rather concise set of symbols which summarize and express this balance between efficiency and self-restraint. First, when a hunter who normally kills perhaps fifty geese in a season suddenly kills, let us suppose, three times that number, it is taken as a sign that the hunter has not long to live. It is not surprising, then, that when an individual hunter has accumulated a larger than average kill early in the season, he sometimes stops for the remainder of the season, or leaves a younger and less experienced hunter in the household to bring home the geese. Collective restraint is also exercised after a particularly abundant daily kill has been made, when all hunters on the territory let the geese rest for a day or two.

There is a second symbol for the perils of excessive killing — the albino Canada goose. Such geese must not be killed under any circumstances, or the hunter will find it very difficult to kill geese thereafter. Thirdly, there is the sandhill crane, a relatively uncommon bird on the east coast of James Bay that is most numerous when there is an unusual abundance of migrating geese feeding and resting in the coastal bays. It is permitted to shoot the crane, but it must be perfectly done. To miss a shot is a sign of the hunter's impending death, or that of a near relative. Fourthly, there is a rarely reported Canada goose that is said to bear a luminous collar over its neck and across its breast. The hunter fortunate enough to see this goose must kill it with a single clean shot, and retrieve it almost

the moment it touches the ground. Otherwise, the collar dissipates, passing to another goose. But the impeccable hunter may reach it in time, and retain the collar as a charm. The geese will thereafter fly to such a hunter, even when he is not well-hidden.

To summarize :

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| (1) Too many geese killed (excessive killing) | = Hunter's death |
| (2) Albino goose killed (excessive killing) | = Curtailment of gifts (hunter's poverty) |
| (3) Crane attempted but not killed impeccably | = Hunter's death |
| (Crane not attempted, or attempted but killed impeccably) | = No special consequence |
| (4) Collar-bearing goose killed impeccably | = Abundance of gifts (hunter's wealth) |
| (Goose not killed impeccably) | = No special consequence |

This set of symbols signifies that the hunter must practice excellence in the killing of game if he is to avoid poverty and death. There is such a thing as "insufficient" killing, expressed in its negative form in the sanction of death when the crane is attempted but failed, and in its positive form in the reward of food wealth when the collar-bearer is impeccably taken (the opposite of death). But there is also the possibility of "excessive" killing, expressed in a strong negative form in the sanction of death for killing too much game, and in the milder negative form of poverty for killing an albino. Symbols of the necessity for impeccability have "literal" implications where efficiency and excellence in a demanding environment make the difference between prosperity and poverty, life and death. But symbols of the necessity for restraint are equally intelligible in literal terms, because the generosity of animals is empirically exhaustible. Geese hunted too noisily or too often in the same place, too much game killed and too little allowed to escape, could lead objectively to dwindling exchange with hunters, poverty, and (in the not too distant past) death.

Let us now turn to the second of the general propositions cited earlier — that positive reciprocity in human society enhances reciprocity with geese. In what literal sense could interspecies reciprocity depend on human reciprocity? Even more important than sharing food, given households' pride in their autonomy, is sharing the *opportunities* to kill geese or other game. I have already mentioned that the management of a hunting territory is a delicate matter, requiring a cooperative strategy. Cooperation becomes impossible when generosity is not maintained. For example, if a hunting boss "saves" a rich hunting spot, to hunt it for the sole benefit of his own household without notifying other hunters so they may join in, his reputation as a leader is seriously damaged. Other hunters then feel justified in hunting when and where they choose on such a territory, with the possible result that rotational management becomes impractical. The geese become increasingly wary, or move elsewhere, and hunting productivity on the territory declines. The animal gift, in this literal sense, depends quite literally on human generosity, and it is this social knowledge more than any other factor that accounts for restraint and reglementation in hunting. Social and ecological reciprocity are merely formally interdependent, as offspring of the same "root" metaphor or paradigm, but also

by virtue of actors' knowledge of the specific empirical circumstances of human-human and human-animal reciprocity.

Both of the general propositions discussed depend on situational elaboration of the reciprocity metaphor, such that the literal and empirical contexts of geese and hunters are assimilated to the terms of the metaphor. It is not as though reciprocity applies "literally" to the human term but only "metaphorically" to animal term. Neither term can be said to stand as the "primary" meaning of reciprocity. Commonality is assumed, but differences are elaborated and exploited to the last empirical detail, in multiple permutations of the root metaphor across numerous phenomenal domains.

In certain ritual contexts, nonetheless, the hunter merges so radically with the animal that differences are quite beside the point. In dreams, for example, geese are vehicles of the hunter's power and essence, and may save him from sorcery. All his life, the hunter receives the gift of geese, and at the hunter's death, it is often a goose that assumes his soul on its journey from this life. Often, at the time of a hunter's death, a solitary goose flies low overhead, or lands near his relatives, acting quite unafraid. The human subject's experience of animals as interpretants of his essential self renders all the more poignant the inevitable separation of hunter from prey, and all the more compelling the morality which joins them in the reciprocity of life-giving and life-taking. The transcendence of this tension at significant moments — the death of an animal, a dream encounter, or the death of a hunter — is the experience of the sacred.

Conclusions :

The complementarity of figurative and literal aspects of signification can perhaps be pushed for some final yields, hopefully not to the point of dwindling returns. The inclusion of humans in a figurative world of analogous other-than-human persons does appear to promote ecological sensibility, as "human relationships of all kinds come to be 'two-sided' and dialectical rather than 'one-sided', linear and exploitative" (Wagner 1977 : 405). A number of studies (Rappaport 1968, Reichel-Dolmatoff 1976, Feit 1978, Nelson 1983) support the conclusion that the "anthropomorphic" categories and relations of many traditional societies are of a kind and quality that approximate scientific ecological concepts and values (scientific ecology, for Wagner, is itself a dialectical departure of a sort from the lineal and exploitative tendencies of science).

The present analysis suggests that the functional or structural correspondences between native cosmology and scientific ecology originate in the practical knowledge of actors, for whom both figurative and literal practices are deliberate. Relational modelling is crucial in defining and redefining the contexts that are assimilated to *differentiae* in alternative moments of metaphoric practice. It is oscillation between figurative and literal modalities of knowledge construction that comprises socially and ecologically adequate practice for Cree hunters.

Wagner believes that semiotics can help us to transcend our own ecology's dialectical dependence on the literal science of nature, its innate counterpart, because semiotics is "comparatively free of broader symbolic associations in Western social science by dealing with symbolic form rather than symbolic

content" (1977 : 405). This freedom is apposite, however, only insofar as the phenomena of experience are not dismissed as "content". The "continuous birth" of the world turns up surprises which inevitably interrupt both literal relations and figurative alignments. An adaptive semiotic will admit the refractory forms of experience along with the activity of invention, if semiotic and natural worlds are truly to be mutually continent of one another*.

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NOTES

1. The sequence here described in abbreviated fashion is one variation on a general ritual structure for "bringing home animals" (see Tanner 1979).

2. Feit (1973, 1978) has discussed the ecological significance of respect for animal gifts. Animals are felt to be given at times and places in which, by virtue of numerical availability and characteristic behavioural traits, they present themselves to the hunter's weapons or traps with maximum efficiency and minimum struggle. When hunters notice animals becoming scarce and difficult to catch, they say it is because the animals are "angry", perhaps because hunters have been taking too many. The respectful response is to stop hunting the species in question until it once again is more freely given.

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